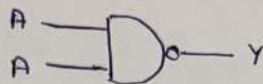


Mission 1:

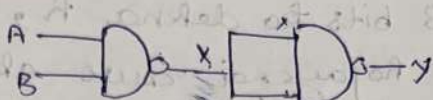
$$\text{NAND} = \overline{A \cdot B}$$

a) A Not gate.



A	Y
0	1
1	0

b) An And gate.



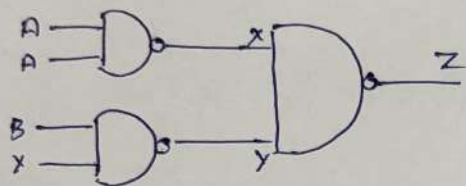
A	B	X	Y
0	0	0	0
0	1	0	0
1	0	0	0
1	1	1	1

Two consecutive NAND gate.

c) An OR Gate

$$\overline{A+B} \Rightarrow \overline{A} \cdot \overline{B}$$

$$A+B = \overline{\overline{A} \cdot \overline{B}}$$



A	B	X	Y	Z
0	0	1	1	0
1	0	0	1	1
1	1	0	0	1
0	1	1	0	1

Mission 2

a)

A	B	S
0	0	0
0	1	1
1	0	1
1	1	0

XOR.

A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

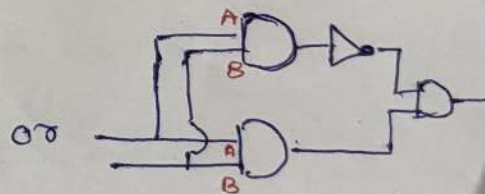
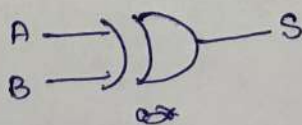
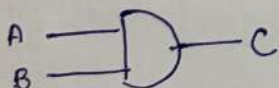
→ AND Gate.

b)

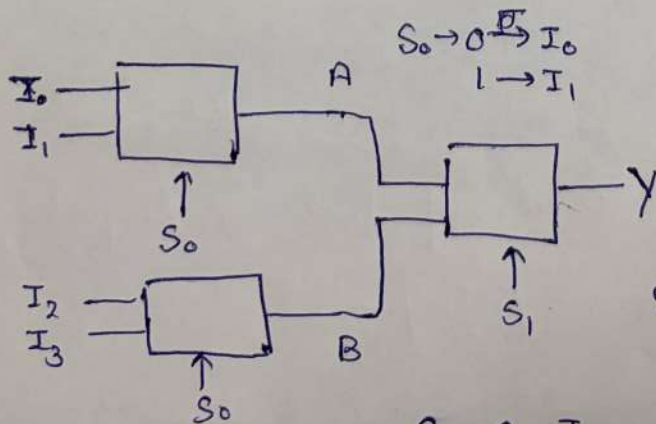
$$S = A \oplus B$$

$$C = A \cdot B$$

c)



Mission 3



$$S_0 = 0 \rightarrow I_0$$

$$1 \rightarrow I_1$$

$$S_1 = 0 \rightarrow A$$

$$1 \rightarrow B$$

$$S_0 = 0 \rightarrow I_2$$

$$1 \rightarrow I_3$$

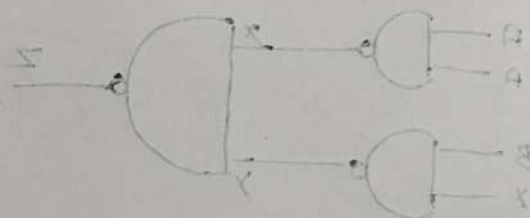
S ₁ S ₀	I ₀
00	I ₀
01	I ₀
10	I ₂
11	I ₃

mission 4: Pattern Detector

a) 2 Flip-Flop gate.

Like Amko 3 bits ko dekhna h, to 2 best input store hojayengi aur ek current input hoga. jb all three is equal to 1 then output 1 aayayega.

X	Y	Z	A
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1



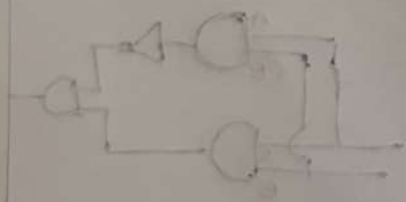
$$\overline{A} \cdot \overline{A} = \overline{A} + \overline{A}$$

$$\overline{A} \cdot \overline{A} = \overline{A} + \overline{A}$$

X	Y	Z	A
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

X	Y	Z	A
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Mission 5 The Counter Mystery.



$$A \oplus A = 0$$

$$A \oplus A = 0$$



A	B
00	00
01	01
10	10
11	11

